XRPL Validator — Daily Engineering Report · 2026-08-03

Period: 2026-08-03 00:00 to 24:00 US Eastern (2026-08-03 04:00 to 2026-08-04 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ledger): the POOL can activate a strand, and an active strand reaps** (`6e631e7`, xrpl-validator) rippled filters strands on `qualityUpperBound` — an OPTIMISTIC bound on what the strand could conceivably do (StrandFlow.h:696-699, `qualityUpperBound(sb, *strand) < *limitQuality => continue`). For a book with an AMM that bound includes the pool. A strand whose every BOOK level is worse than the limit therefore still gets built, its offer stream still steps, and the dead offers it steps past are still reaped — even …
- **feat(ledger): reverse pass — size each payment hop to what the next needs** (`9ac6bcc`, xrpl-validator) rippled flows a strand in TWO passes: a REVERSE pass walking back from the requested output working out how much input each step needs (StrandFlow.h:105-230, `for (auto i = s; i--;) strand[i]->rev(...)`, re-executing a limiting step when one binds), then a FORWARD pass bounded by what is actually available. Ours was forward-only, so an intermediate hop sold its WHOLE carry rather than buying what the hop after it …
- **feat(ffi): capture libxrpl's decision trace under XRPL_FFI_TRACE** (`8b5a197`, xrpl-validator) The shim already had a CapturingSink wired into xrpl_apply_with_mutations and surfaced through xrpl_result_last_fatal — but gated at Severity::Fatal, so everything rippled narrates was thrown away. Lift it to Trace when XRPL_FFI_TRACE is set, and dump it per transaction with XRPL_FFI_TRACE_TX=. Both OFF by default: gate runs stay byte-identical and trace capture is neither free nor small.
- **fix(ledger): round a fill's XRP input by which side bounds it** (`d6f7589`, xrpl-validator) The taker's input was always ceiled to whole drops. rippled's direction depends on which side of the fill is fixed — its rev/fwd split: - the taker's remaining WANT bounds it: the input is computed FROM that output (`rev`) and rounds UP, so the taker pays enough; - the maker's FUNDING bounds it: the output is what is fixed and the input follows it (`fwd`), rounding DOWN.
- **fix(ledger): judge a fill's achieved quality exactly, as rippled does** (`133fd89`, xrpl-validator) rippled discards a pass outright when the quality it actually realised misses the limit: if (limitQuality && q < *limitQuality && (!adjustedRemOut || !withinRelativeDistance(q,* limitQuality, Number(1, -7))))

- **fix(ledger): one AMM turn per pass, and a rejected pass ends the crossing** (`8152aca`, xrpl-validator) Three pieces, each only visible once the previous one landed, all read off libxrpl's own trace rather than inferred.

Independence program

No independence record for UTC day 2026-08-03; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **DID-Linked On-Ledger Reserve Attestations** (audit) Add a new owner-reserved ltRESERVE_ATTESTATION ledger object at keylet::reserveAttestation(issuer, assetID), creatable/updatable only via a new Reserv
- **Confidential MPT Balances, Issuer-Clawback Preserved** (privacy) Add an lsfMPTConfidential flag to MPTokenIssuance and replace the plain integer balance on MPToken with a Pedersen commitment sfBalanceCommitment when
- **XChain Bridge Settlement Finality Certificates** (interop) Add a new protocol-minted (no owner reserve) ltBRIDGE_FINALITY_CERT ledger object at keylet::bridgeFinalityCert(bridgeID, xChainClaimID), created insi

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-03 05:26	—	xrpl-validator	`6e631e7`	fix(ledger): the POOL can activate a strand, and an active strand reaps
08-03 05:27	—	xrpl-validator	`9ac6bcc`	feat(ledger): reverse pass — size each payment hop to what the next needs
08-03 14:21	—	xrpl-validator	`8b5a197`	feat(ffi): capture libxrpl's decision trace under XRPL_FFI_TRACE
08-03 15:22	—	xrpl-validator	`d6f7589`	fix(ledger): round a fill's XRP input by which side bounds it
08-03 16:09	—	xrpl-validator	`133fd89`	fix(ledger): judge a fill's achieved quality exactly, as rippled does
08-03 17:11	—	xrpl-validator	`8152aca`	fix(ledger): one AMM turn per pass, and a rejected pass ends the crossing

Times in UTC. A commit appears on the day it was committed to main or merged into it.